

- 1 (a) (i) State the base unit of temperature.
- (ii) Boltzmann's constant, k , and Planck's constant, h , are related to Stefan's constant, δ , by the expression

$$\delta = \frac{ak^4}{h^3c^2}$$

where a is a dimensionless constant and c is the speed of light

Determine the base units of δ .

- (iii) State the limitation of using base units in the equation in (ii). [5]

- (b) (i) Define the terms *accuracy* and *precision*.

- (ii) State and explain the error associated with precision.

- (iii) The following results in Nkg^{-1} were obtained in an experiment to determine the earth's gravitational field strength, g

8.6, 8.7, 8.8, 8.5, 8.2, 8.9

Discuss whether the results were precise and/or accurate.

[6]

- (c) Fig.1.1 shows part of a car moving at 15 ms^{-1} on a level road. The wheel is of radius 0.30 m and has an angular velocity, ω , of 50 rad s^{-1} .

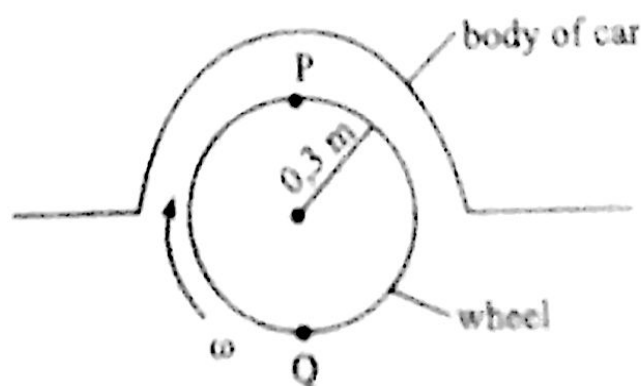


Fig.1.1

- (i) Determine the minimum coefficient of friction required for the car to move without skidding.

- (ii) Point P is observed as moving at twice the velocity of the centre of the wheel while point Q is observed as momentarily at rest.

Explain these observations.

[5]

- (d) The system shown in Fig.1.2 is in equilibrium.

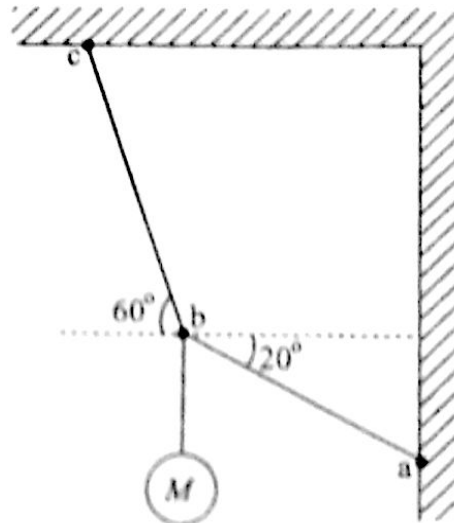


Fig.1.2

If $M = 2.0 \text{ kg}$, find the magnitude of the tension in

- (i) string ab,
(ii) string bc

[4]

2

- (a) Distinguish between a vector quantity and a scalar quantity.

[2]

- (b) A car travelling at 20 ms^{-1} due East turns and continues at 15 ms^{-1} due South.

- (i) Draw a vector diagram to illustrate the

1. initial and the final velocity,
2. change in velocity.

- (ii) Calculate the change in the

1. speed of the car,
2. velocity of the car

- (c) Explain why the acceleration of an object may not be in the same direction as the velocity of the object. [2]

- 3 (a) (i) Define a couple. [1]
 (ii) State two conditions for a body to be in equilibrium. [3]

- (b) Fig. 3.1 shows a horizontal uniform beam of length 8 m and weight 200 N. The beam is pivoted at a wall with one of its ends connected to a metal cable of an angle of depression of 53° . A mass weighing 600 N is placed 2 m from the wall. A reaction force, R , acts at the pivot, P , at an angle, θ , to the horizontal.

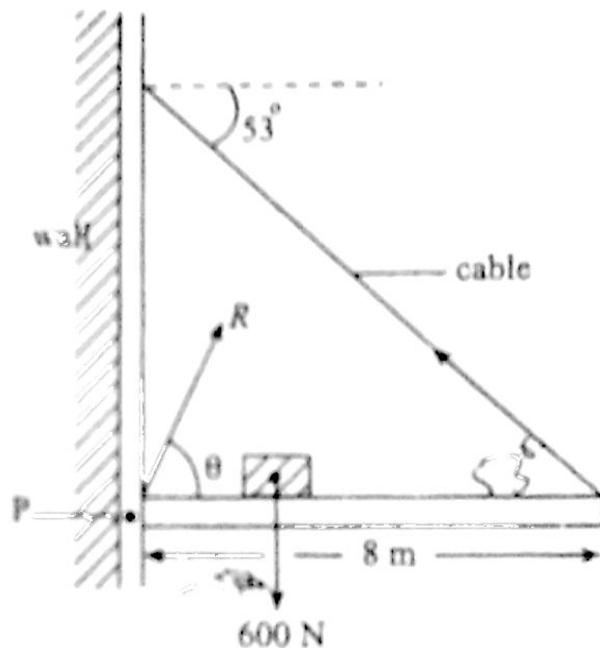


Fig. 3.1

Calculate the

- (i) tension in the cable,
 (ii) reaction force, R ,
 (iii) angle θ .

[6]

- (c) Give one example of everyday application of forces in equilibrium. [1]

- 4 (a) Define gravitational field strength. [1]

- (b) A satellite is in a circular orbit of radius, r , about the Earth of mass, M .

- (i) Deduce an expression for the period of the satellite.

- (ii) A satellite of mass 500 kg is in circular orbit at a height of 2 000 km above the earth's surface. The radius of the earth is 6 400 km.

Calculate the

1. gravitational field strength at the height of 2 000 km,
2. orbital speed of the satellite.

[7]

- (c) Suggest why a geostationary satellite has a short lifespan.

[2]